

Fake News Detection

A Hybrid Stylometry & Transformer Approach

Group 13

Advanced Machine Learning | UZH



Phase 1: Formal problem setting & dataset scope

Objectives

Supervised binary classification over news text, enriched with linguistic markers for emotionality and readability.

Task definition

Input: article token sequence x and feature vector z $\phi(x)$

Output: label $y \in \{0,1\}$; 1 = fake, 0 = real

Fake-News-Detection Datasets From: [Kaggle](#)

Fake News: 23.5k

2015 - 2018

News 39%, Politics 29% and Other 32%

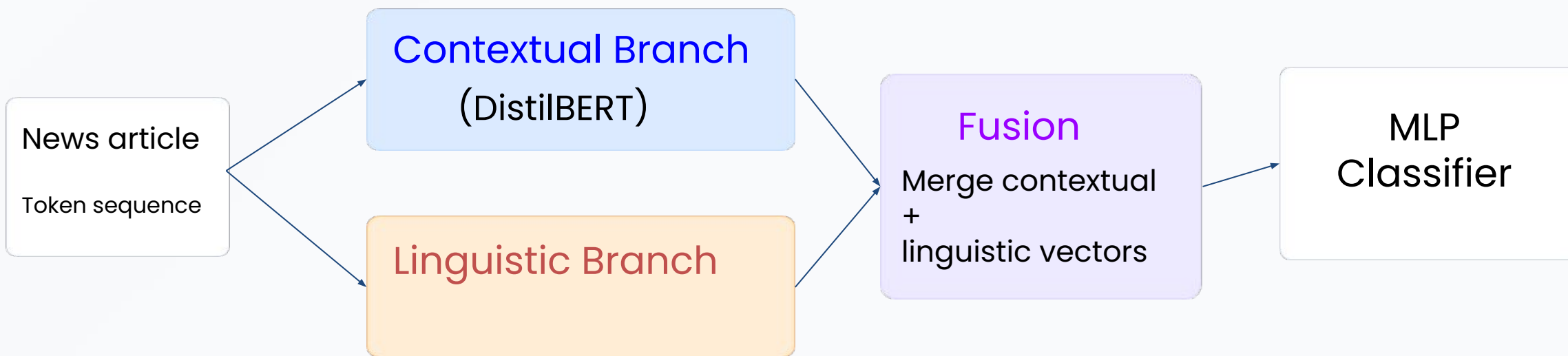
True News: 21.4k

2016 - 2017

PoliticsNews 53%, worldnews 47%

Proposed Approach: Hybrid Fusion Architecture

Key Idea: combine DistilBERT contextual embeddings with sentiment polarity and Gunning Fog Index features.



> More about this after we introduce the baselines.

Training, Tuning & Evaluation Protocol

Data split

Train 70%	Dev 15%	Test 15%
Train: 31,428	Dev: 6,735	Test: 6,735

Hyperparameter search

- ❑ Tune on Dev set only
- ❑ Learning rate grid: 0.01, 0.001, 0.0001
- ❑ Dropout: 0, 0.1, 0.5
- ❑ (Batch size: 8, 16)
- ❑ Using:
hyperparameter_search() and optuna
- ❑ (optimizing evaluation loss)

Metric focus

- ❑ Primary: AUROC
- ❑ Optimize: Precision
- ❑ Monitor: F1-Score

Rationale: catch fake news while avoiding false accusations of real news as fake.

Benchmark

- ❑ Hybrid model: DistilBERT + Linguistic features
- ❑ Vanilla DistilBERT: no parameter tuning and without feature injection
- ❑ Random Forest + TF-IDF: statistical baseline

Final test set is used once for unbiased evaluation after tuning.

The 4 Linguistic Indicators



Sentiment Polarity (TextBlob)

A baseline emotional score ranging from **-1.0 (highly negative)** to **+1.0 (highly positive)** based on a static word dictionary.



Compound Sentiment (VADER)

A nuanced emotional score (also from **-1.0 to +1.0**) that is specifically optimized for social media text. It accounts for capital letters, emojis, and exclamation marks.



Text Complexity (Gunning Fog Index)

The readability of the text expressed as a U.S. school grade level. It analyzes average sentence length and the percentage of complex (multi-syllable) words.



Lexical Diversity (Type-Token Ratio)

Vocabulary richness. It divides the number of unique words by the total number of words. Lower diversity means the text repeats the same words over and over.

Phase 2: Training and Evaluating the Baselines

The "Cheatcode" Crisis

The Spider-man Reality

In our first run, accuracy hit 99.9%. We realized the model was just checking for the agency tag "**Reuters**".

The Fix: `linguistic_features_aggressive.py` strips all formatting to force the model to learn true stylometry.

As such, the more aggressive clean involves:

- stripping "Reuter" and variations
- lower casing everything
- removing article section (eg Global News etc)
- removing year of publication
- stripping away extra spaces which often giveaway fake news



Baseline 1: Random Forest + TF-IDF

0.979

F1-Score (Test)

Feature Mania

These results are generated by using the 4 custom features as well as 10,000 max feature setting for the TfidfVectorizer function.

Baseline 1.5: Text-Agnostic RF

0.708

F1-Score (Test)

Pure Stylometry Power

Results using strictly the 4 engineered features. This proves stylistic patterns exist even without context.

$$F_1 = 2 \cdot \frac{P \cdot R}{P + R}$$

Baseline 2: Vanilla DistilBERT

0.997

F1-Score (Test)

DistilBERT with no custom features

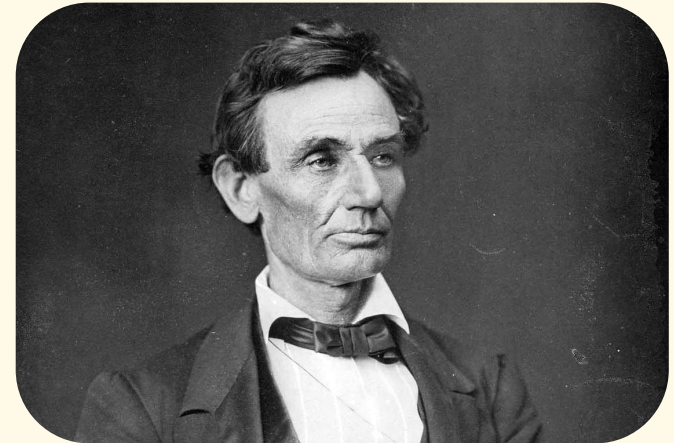
Results using the hugging face distilbert model and training it on 3 epochs with some random/default parameter settings. The cleaned up version of the data is used for training and evaluation.

The Authentication Dilemma

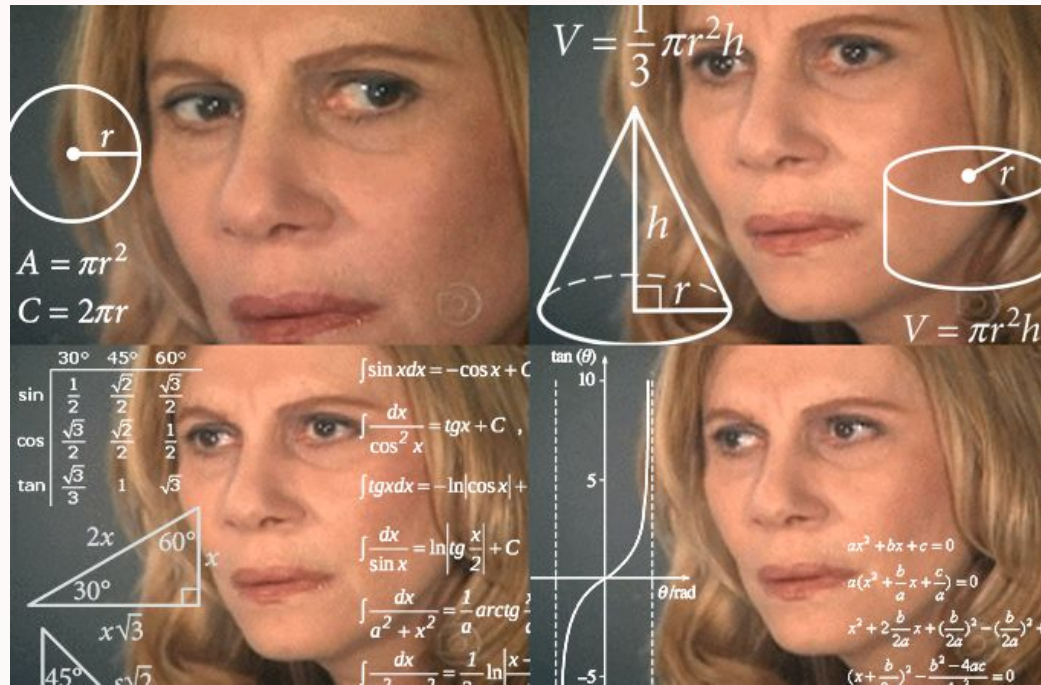
"Don't believe everything you read on the internet just because there's a picture with a quote next to it."

— Abraham Lincoln (1864)

(This proves why semantic context must be verified by structural style)



DistilBERT Context Power



Semantic Mastery

We use DistilBERT for its balance of efficiency and depth. It captures the semantic "flavor" of misinformation.

- ⚡ 40% Fewer parameters.
- ⚡ 60% Faster training.
- ⚡ 99% Test F1-Score.

Phase 3: Training and Evaluating the Hybrid Architecture

The Hybrid Architecture

The model learns when to trust semantic context vs. when to prioritize structural fingerprints.

$$\text{Output} = (G \times T) + ((1 - G) \times L) (*)$$

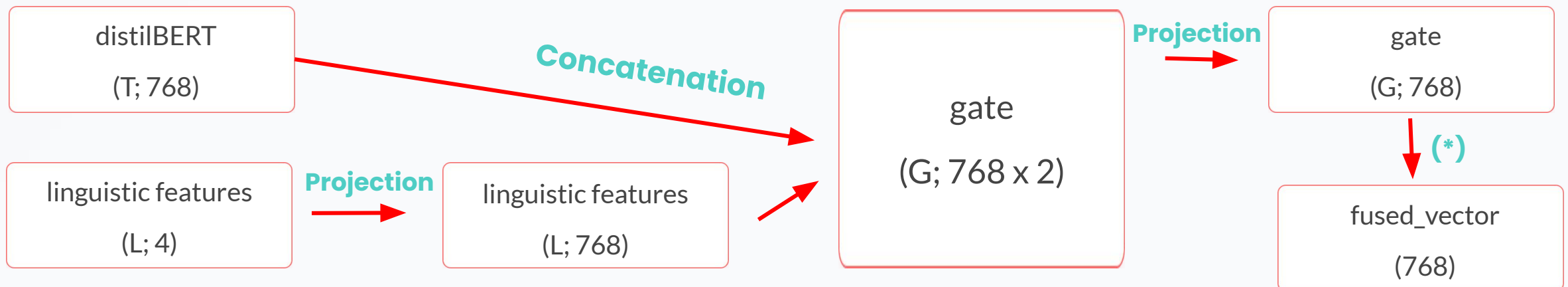
T = text vector (distilBERT)

L = linguistic vector

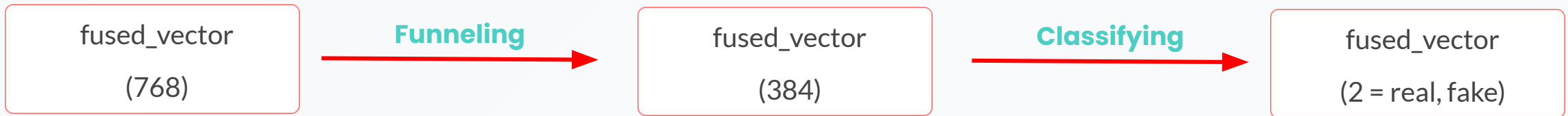
G = gate

Finding G

The model uses a neural network layer that looks at the text and the writing style at the same time, turning them into a vector of percentages via projection.



The Hybrid Architecture



Example:

Gate value = 0.9 => trust the text 90% and linguistic features 10%
= 0.5 => trust the text and linguistic features in equal measures
= 0.1 => trust the text only a little and focus on linguistic features

Reasoning: For sensationalist articles, the gate should lean heavily on linguistic sentiment because the text embedding alone may not capture exaggeration and 'click-bait-y' elements well enough

fused_vector = (3.4, 1.7) => Classify the article as "real" since $3.4 > 1.7$

Final Performance Comparison

Model Configuration	F1-Score	AUC-ROC	Accuracy
Agnostic Baseline (Style Only)	0.7081	0.7453	0.6742
Baseline RF + TF-IDF	0.9794	0.9972	0.9786
Gated Hybrid Model	0.9919	0.9988	0.9916
Vanilla DistilBERT	0.9971	0.9999	0.9997

Questions?

Thank you for your attention!

Group 13: UZH AML Project

Repo: [LeaC0d3s/aml2026-group-13](https://github.com/LeaC0d3s/aml2026-group-13)